Xamarin.Forms 5 and The Future of .NET

David Ortinau | @davidortinau

February 9, 2021





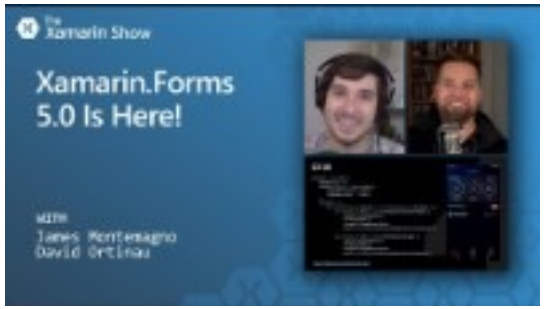

David Ortinau
Principal Program Manager at Microsoft
Session:
A .NET MAUI Progress Report
.NET Frontend Day
www.dotnet-frontend.com


DotNet 2020
Xamarin.Forms 5
David Ortinau


MonkeyFest USA
.NET MAUI Handlers
Javier Suárez

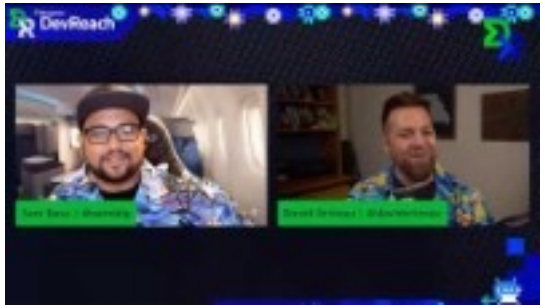

Introducing .NET MAUI
Xam Expert Day
David Ortinau
2020



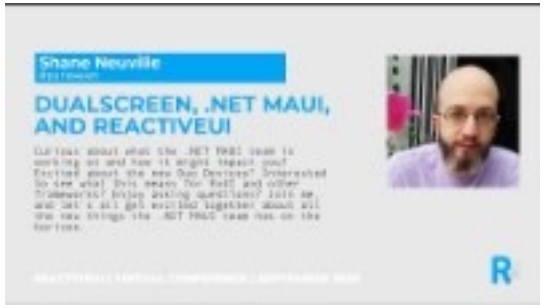

The Xamarin Show
Xamarin.Forms 5.0 Is Here!
with
James Montemagno
David Ortinau


Nice to meet you
Xamarin Community Toolkit Introduction


MonkeyFest USA
Xamarin in .NET 6
Jonathan Peppers


DevReach

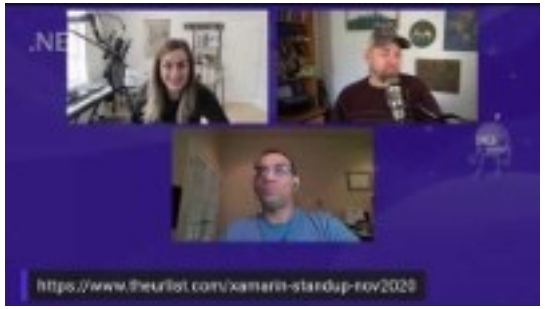

.NET MAUI MVU
Xam Expert Day
James Clancey
2020


Shane Neuville
DUALSCREEN, .NET MAUI, AND REACTIVEUI


Maddy Leger
What's new with Xamarin?


What's New & Next for Xamarin Developers


DOCTORS HATE THIS!
5 Things About Xamarin.Forms 5


https://www.theurlist.com/xamarin-standup-nov2020

The New .NET Multi-platform App UI



David

February 3rd, 2021

With .NET 6 previews starting right around the corner, it is time to start getting excited for the new .NET Multi-platform App UI (MAUI) that was announced at BUILD 2020. This year of .NET has a lot of amazing things for client application developers. Let's review the highlights and set your expectations for the year ahead.

.NET 6 planning and execution is happening in the open, and we have a beautiful website that summarizes our GitHub epics and user stories across all of .NET. Visit themesof.net and search on Xamarin to see the high-level details.

Xamarin.Forms 5

David Ortinau
Principal Program Manager
david.ortinau@microsoft.com

Microsoft

Why did you do this instead of that?

🤦‍♀️ Feedback Themes 🤦‍♂️

- Stability
  - Feature completeness
  - Issue resolution
  - Remove experimental flags
- Design
  - Gradients, shapes, templates
- Productivity
  - Less platform code needed







Last Year+ in Xamarin.Forms

- Accessibility
- BindableLayout EmptyView
- Brushes (Solid & Gradient)
- C# UI Extensions
- CarouselView
- Character Spacing
- CheckBox
- Drag and drop
- EmbeddedFonts
- Expander
- FontImageSource
- GIF Animations
- HTML Label
- IndicatorView
- Maps+
- MediaElement
- Modal Prompts
- Multi-Binding
- RadioButton
- RefreshView
- RelativeSource Binding
- Shapes, Paths, Clipping
- Shell Modals
- SourceLink
- SwipeView
- VisualStateManager Target
- WKWebView
- + Much More!

“Experimental” in Xamarin.Forms

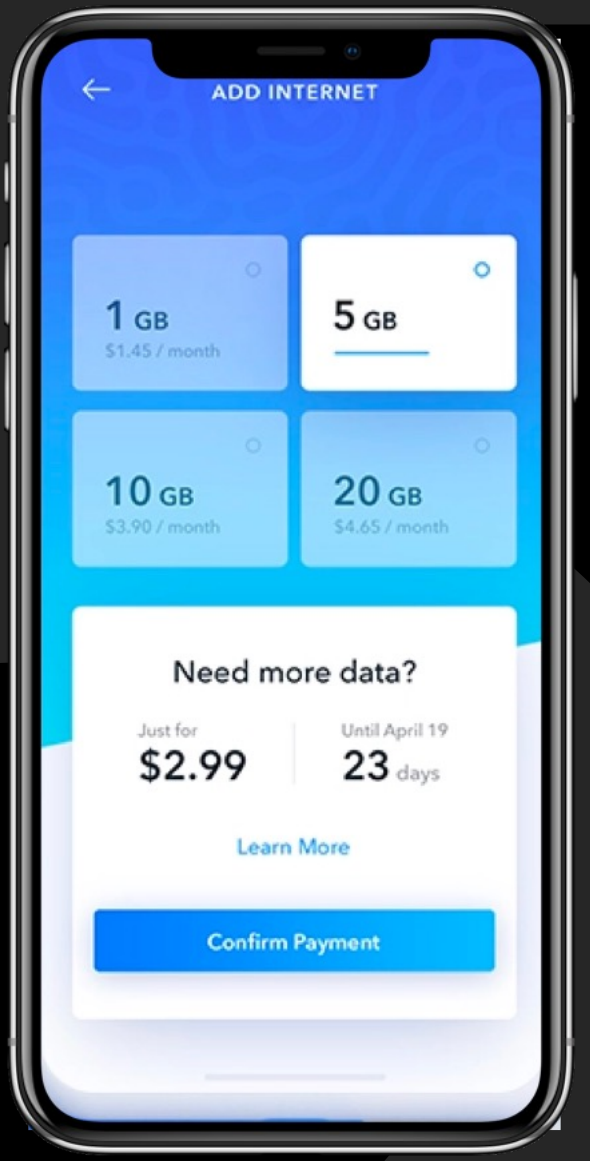
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Brushes

```
<LinearGradientBrush
    x:Key="HorizontalLinearGradient"
    StartPoint="0, 0"
    EndPoint="1, 0">
    <LinearGradientBrush.GradientStops>
        <GradientStop Color="#1C7EFF" Offset="0.1" />
        <GradientStop Color="#3771F7" Offset="0.6" />
        <GradientStop Color="#3877F6" Offset="1.0" />
    </LinearGradientBrush.GradientStops>
</LinearGradientBrush>

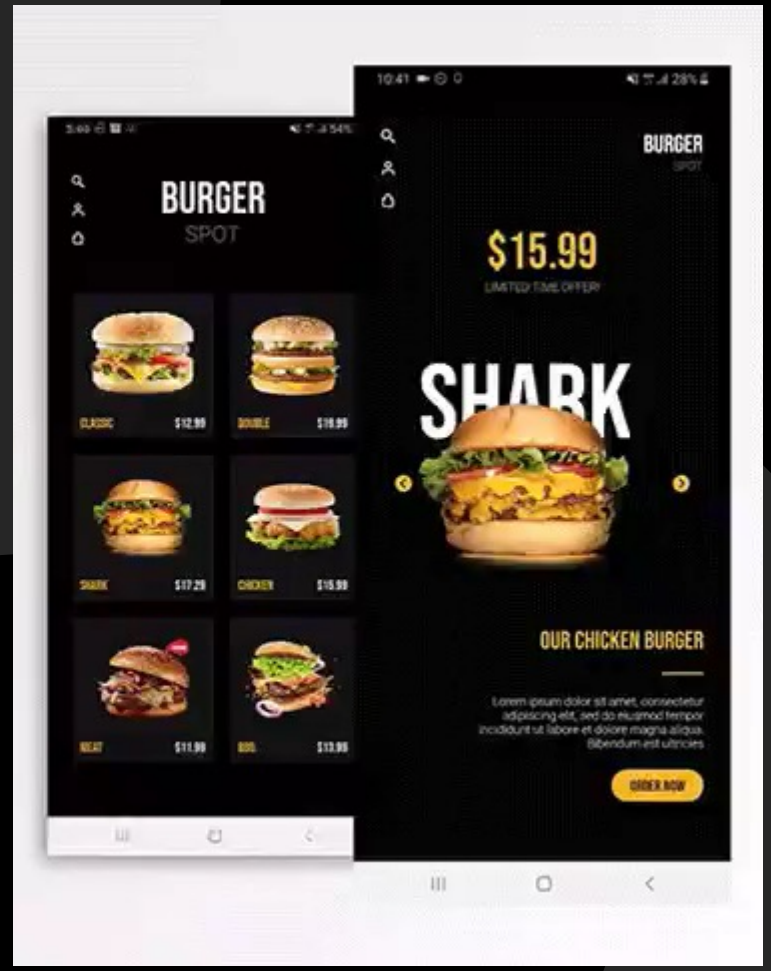

<NavigationPage
    BarBackground="{StaticResource HorizontalLinearGradient}">

</NavigationPage>
```



CarouselView

```
<CarouselView
    ItemsSource="{Binding Burgers}"
    HeightRequest="260"
    HorizontalOptions="FillAndExpand"
    CurrentItem="{Binding SelectedBurger}"
    IsScrollAnimated="True"
    IsSwipeEnabled="True">
    <CarouselView.ItemTemplate>
        <DataTemplate>
            <t:BurgerItem />
        </DataTemplate>
    </CarouselView.ItemTemplate>
</CarouselView>
```

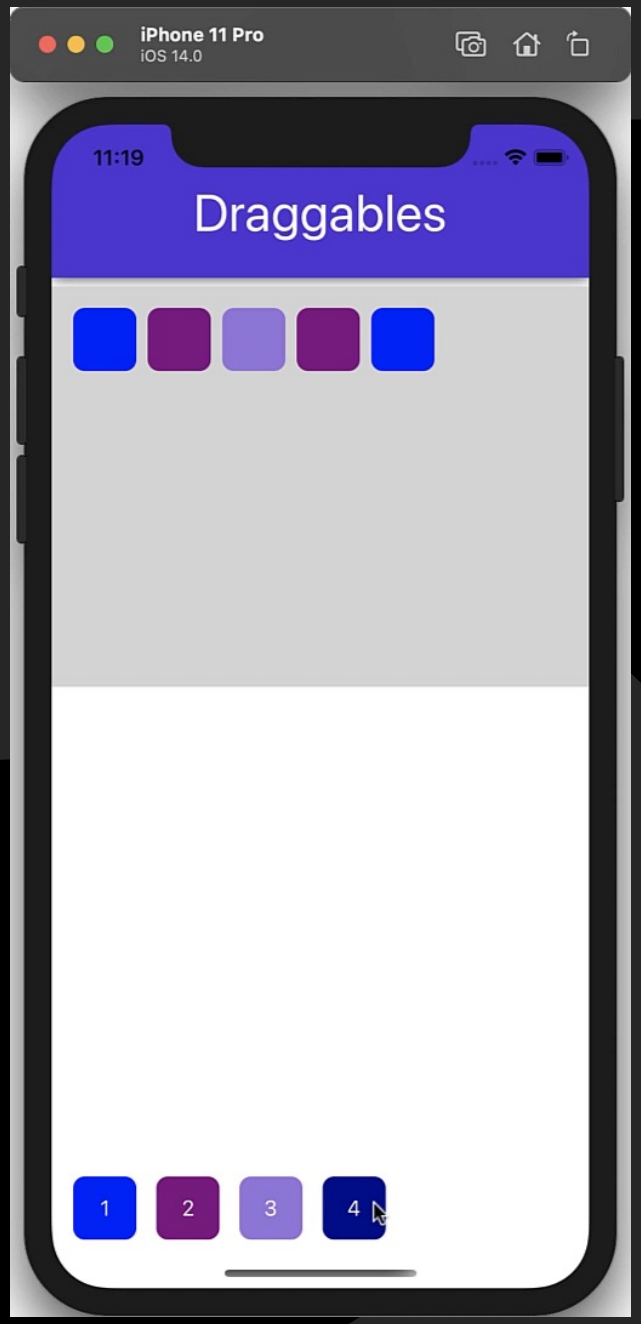


https://github.com/devcrux

Drag and drop

```
<Button
    Text="1"
    BackgroundColor="Blue">
    <Button.GestureRecognizers>
        <DragGestureRecognizer
            DragStarting="DragGestureRecognizer_DragStarting"
            CanDrag="True" />
    </Button.GestureRecognizers>
</Button>

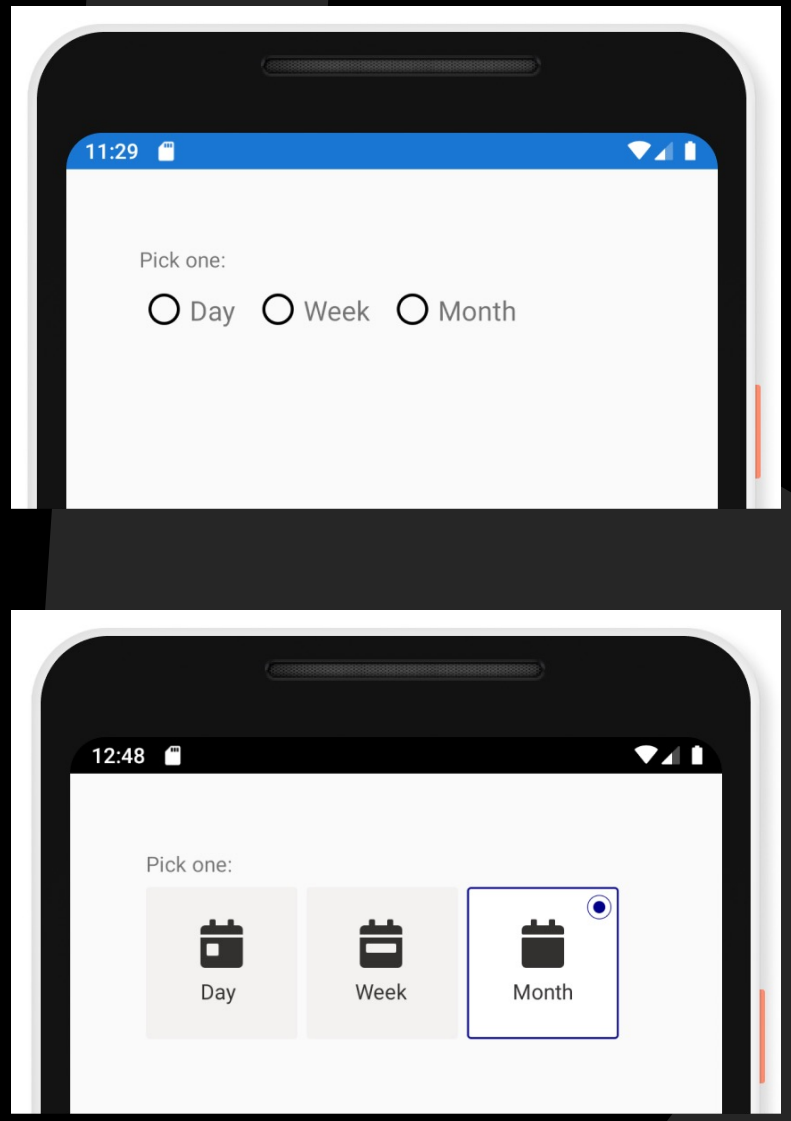
<StackLayout>
    <StackLayout.GestureRecognizers>
        <DropGestureRecognizer AllowDrop="True"
            DragOver="DropGestureRecognizer_DragOver"
            DragLeave="DropGestureRecognizer_DragLeave"
            Drop="DropGestureRecognizer_Drop"/>
    </StackLayout.GestureRecognizers>
</StackLayout>
```



RadioButton

```
<StackLayout
    RadioButtonGroup.GroupName="SimpleRadios"
    Orientation="Horizontal">
    <RadioButton Content="Day" />
    <RadioButton Content="Week" />
    <RadioButton Content="Month" />
</StackLayout>

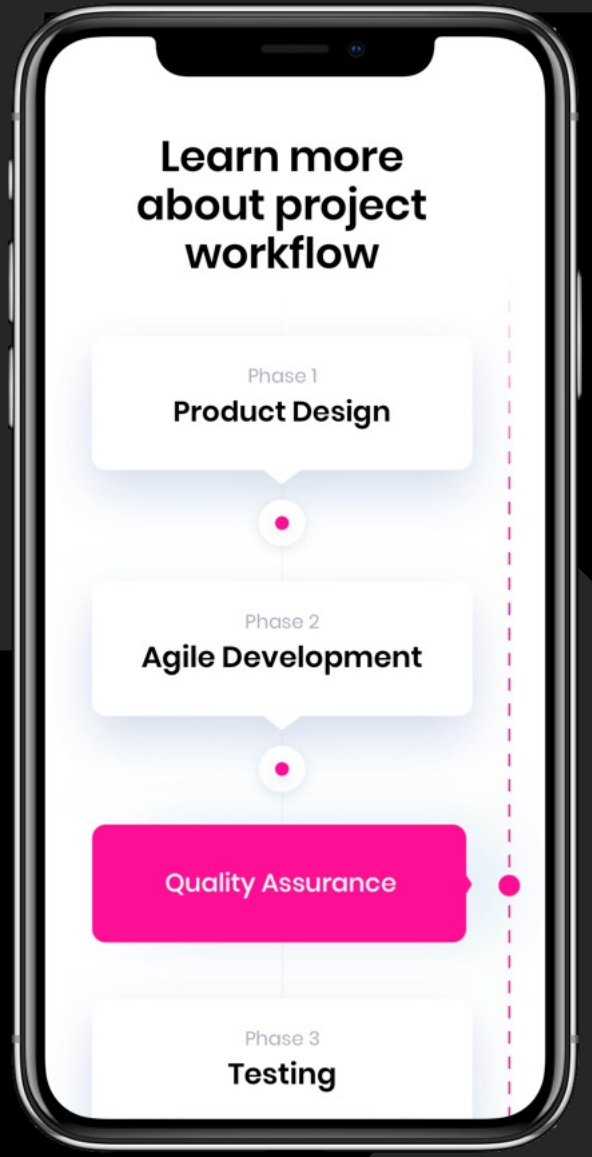
<RadioButton
    ControlTemplate="{StaticResource
                      CalendarRadioTemplate}">
    <RadioButton.Content>
        <StackLayout
            HorizontalOptions="Center"
            VerticalOptions="Center">
            <Image
                Source="{FontImage FontFamily=FontAwesome,
                Glyph={x:Static local:IconFont.CalendarDay},
                      Color=#323130,Size=32}" />
            <Label Text="Day" TextColor="#323130" />
        </StackLayout>
    </RadioButton.Content>
</RadioButton>
```



Shapes & Paths

```
<!— Lines and Dots —>
<Line X1="0" X2 ="300" Stroke="Pink"
    StrokeThickness="2" StrokeDashArray="1,1"/>
<Ellipse HeightRequest="12" WidthRequest="360" Stroke="Pink" />

<!— Conversation Bubble —>
<Grid Margin="12">
  <BoxView
    Grid.RowSpan="2"
    Fill="{StaticResource AccentBrush}"/>
  <Label Text="{Binding Text}">
  <Label
    Grid.Row="1"
    Text="{Binding =Timestamp}"/>
  <Path Grid.Row="2"
    Data="m 0,0 l 16,0 l 0,16 l -16,-16"
    Fill="{StaticResource AccentBrush}"
    Margin="0,0,5,0" />
</Grid>
```



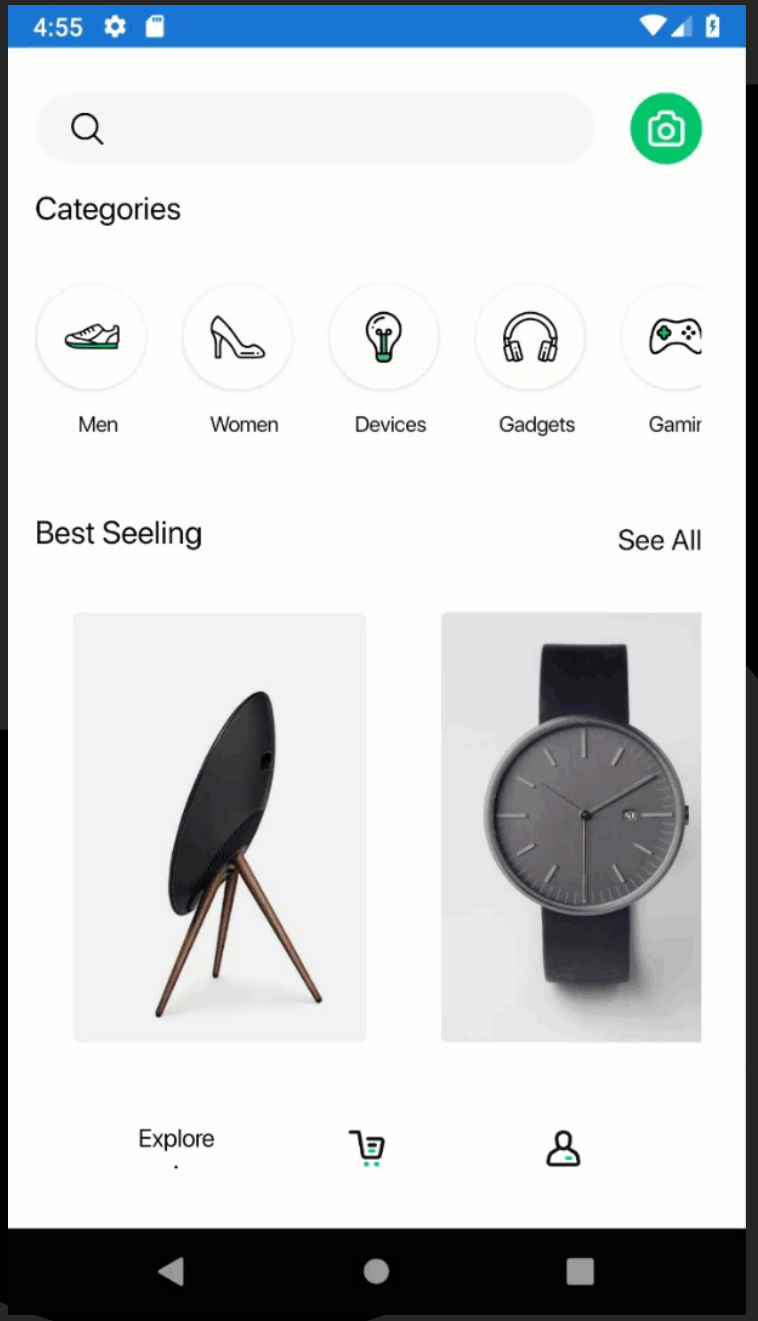
SwipeView

```
<SwipeView>
    <SwipeView.LeftItems>
        <SwipeItems>
            <SwipeItem
                BackgroundColor="#FFC107"
                Invoked="OnFavoriteSwipeItemInvoked"
                Text="Favorite" />
        </SwipeItems>
    </SwipeView.LeftItems>
    <SwipeView.RightItems>
        <SwipeItems>
            <SwipeItem
                BackgroundColor="#FF3D00"
                Command="{Binding DeleteCommand}"
                CommandParameter="{Binding}"
                Text="Delete" />
        </SwipeItems>
    </SwipeView.RightItems>

    <!--  Content  -->

</SwipeView>
```

https://github.com/exendahal



“Experimental” in Xamarin.Forms

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- C# UI Extensions
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- Character Spacing
- CheckBox
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- EmbeddedFonts
- Expander
- FontImageSource
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- + Much More!

Xamarin Community Toolkit

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Xamarin Community Toolkit



Behaviors

- Animations
- Validators
- EventToCommand
- ImpliedOrderGrid
- Masked (input)
- UserStoppedTyping

Converters

- ByteArrayToImageSource
- ItemSelectedEventArgs
- MultiConverterParameter
- NotEqual
- TextCase
- MANY MORE!

Effects

- SafeArea

Extensions

- ImageResource
- Translate
- ValueConverter

Helpers

- Localization
- Subscription
- WeakEvent

ObjectModel

- AsyncCommand
- ObservableRangeCollection

Views

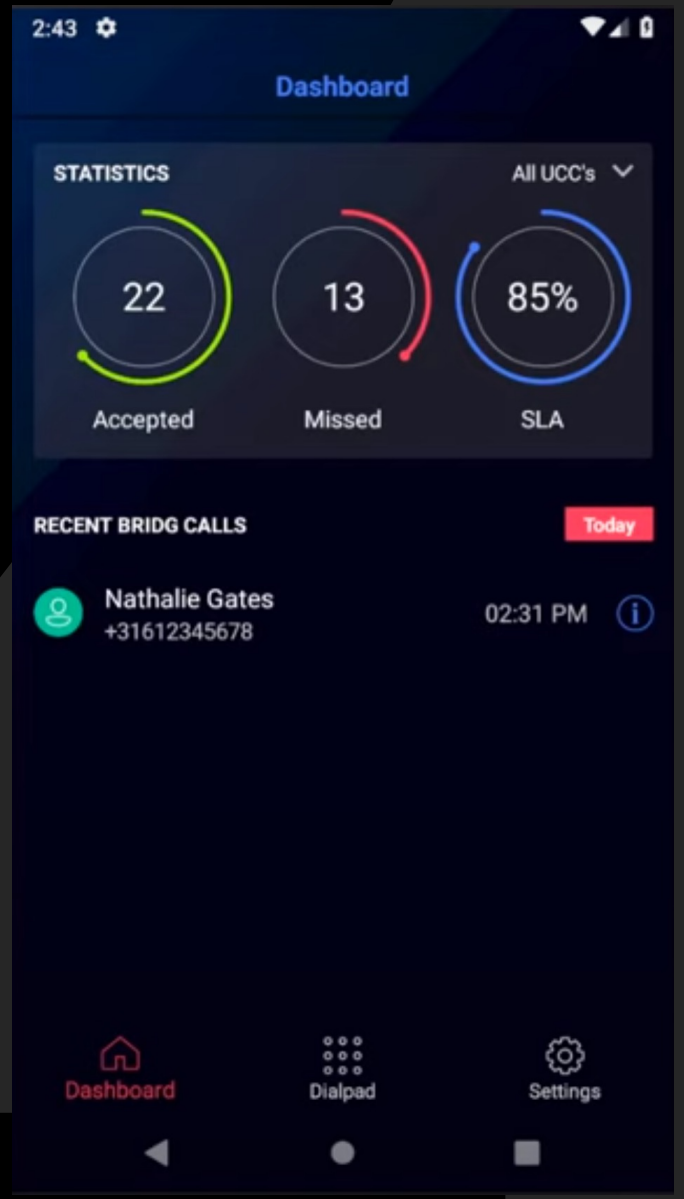
- CameraView
- Expander
- SideMenuView
- AvatarView
- RangeSlider

github.com/xamarin/xamarincommunitytoolkit

C# UI

```
Content = new Grid {
    RowSpacing = 0,
    RowDefinitions = Rows.Define(
        (PageRow.Header, Auto),
        (PageRow.Body  , Star)
    ),
    Children = {
        new ScrollView { Content = new StackLayout { Children = {
            new Button { Text = nameof(RegistrationCodePage) }
                .Style (FilledButton)
                .FillExpandH ()
                .Margin (PageMarginSize)
                .Bind (nameof(vm.ContinueToRegistrationCommand)),

            new Button { Text = nameof(NestedListPage) }
                .Style (FilledButton)
                .FillExpandH ()
                .Margin (PageMarginSize)
                .Bind (nameof(vm.ContinueToNestedListCommand)),
```



https://github.com/VincentH-Net

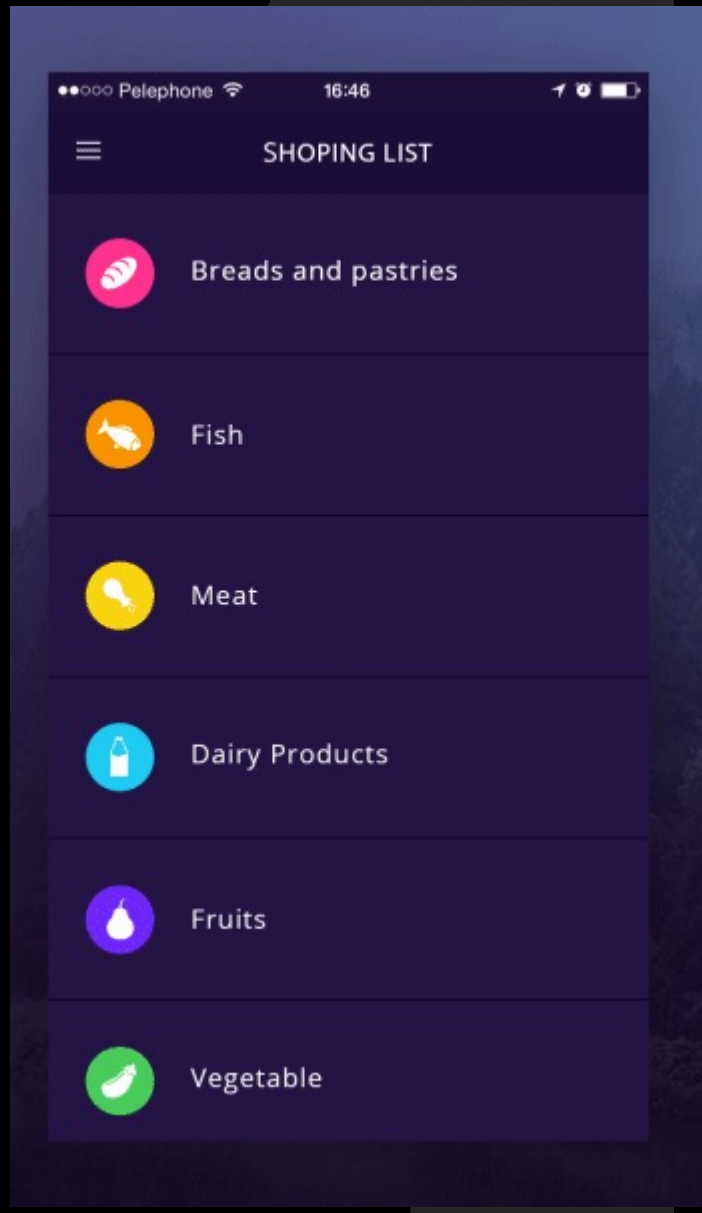
Expander

```
<xct:Expander
    ExpandAnimationEasing="{x:Static Easing.Linear}"
    CollapseAnimationEasing="{x:Static Easing.Linear}"
    IsExpanded="{Binding IsDetailVisible, Mode=TwoWay}">
    <xct:Expander.Header>
        <v:HeaderView />
    </xct:Expander.Header>

    <!-- content -->

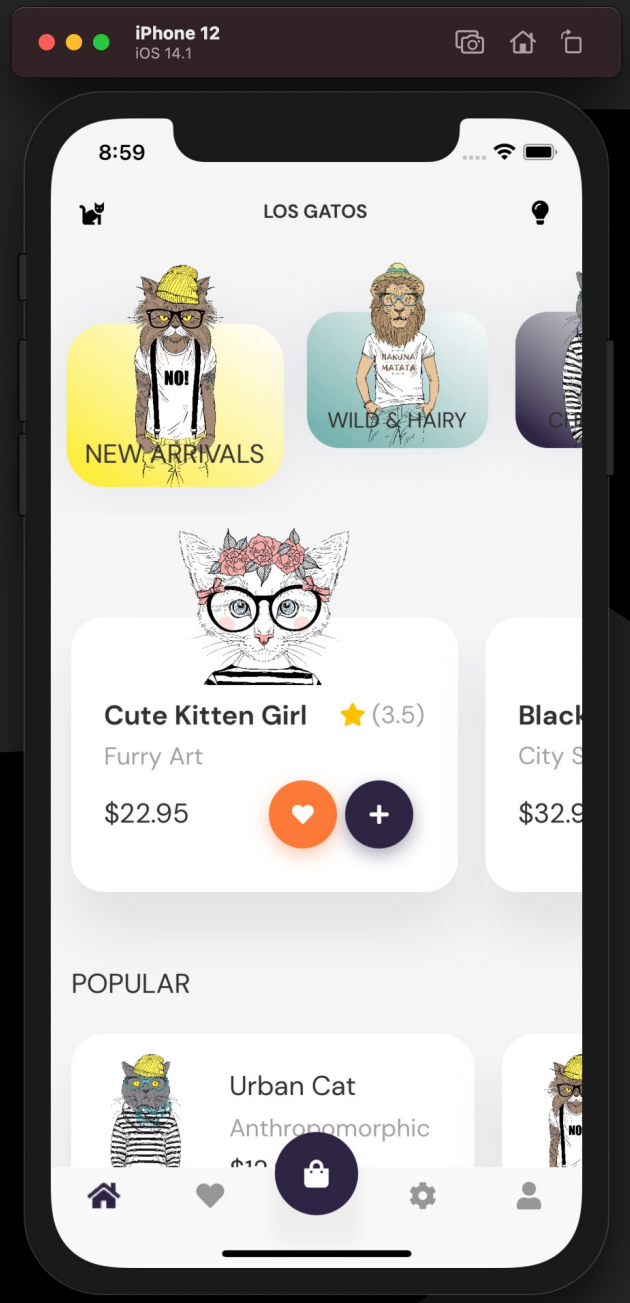
</xct:Expander>
```

https://github.com/jsuarezruiz



TabView

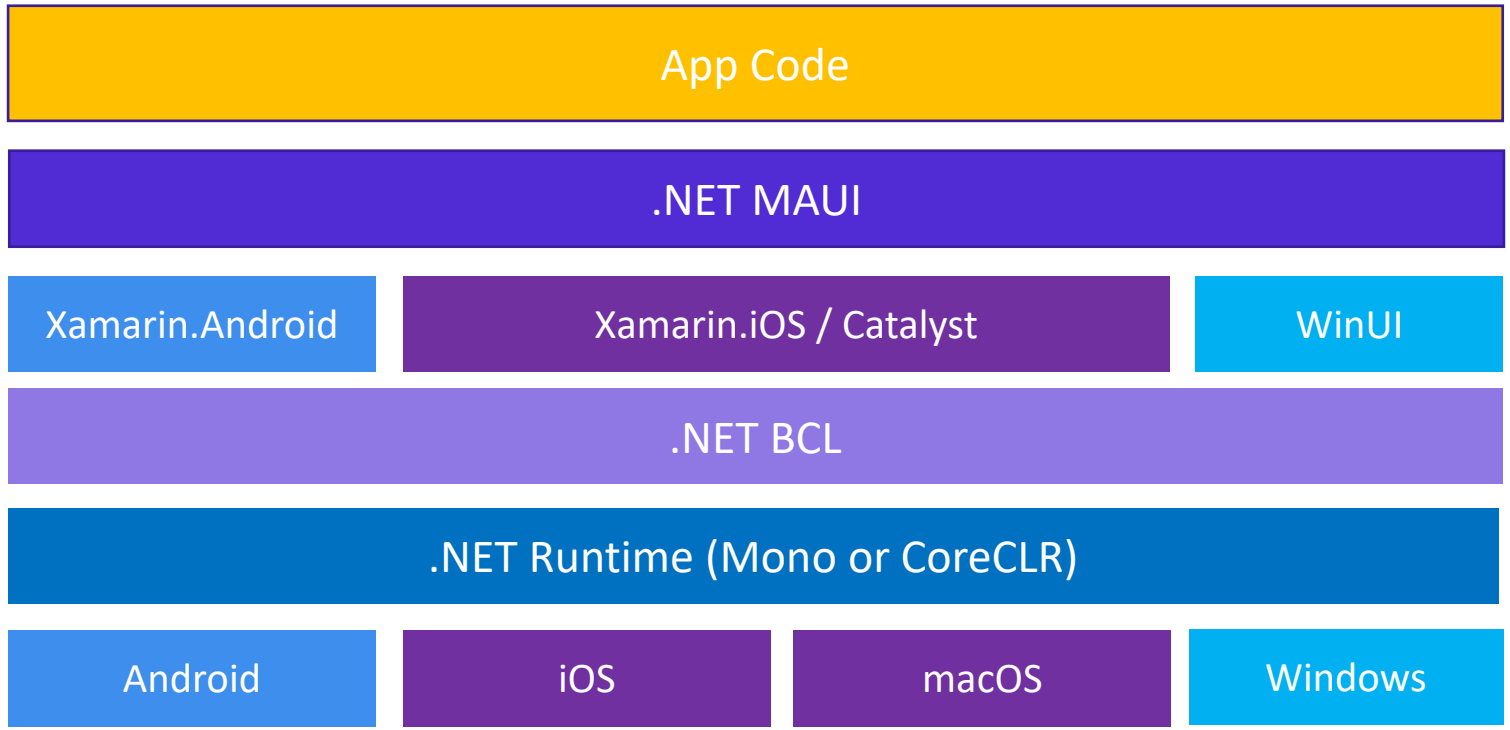
```
<xct:TabView
    Style="{StaticResource CustomTabStyle}">
    <xct:TabViewItem
        ControlTemplate="{StaticResource FabTabItemTemplate}"
        Style="{StaticResource TabItemStyle}"
        Icon="{StaticResource PlusIcon}"
        Command="{Binding AddCommand}" />
    <xct:TabViewItem
        ControlTemplate="{StaticResource TabItemTemplate}"
        Style="{StaticResource TabItemStyle}"
        Icon="{StaticResource ClockIcon}"
        ContentTemplate="{DataTemplate ClockView}" />
    <xct:TabViewItem
        ControlTemplate="{StaticResource TabItemTemplate}"
        Style="{StaticResource TabItemStyle}"
        Icon="{StaticResource AlertIcon}"
        ContentTemplate="{DataTemplate AlertView}" />
    <xct:TabViewItem
        ControlTemplate="{StaticResource TabItemTemplate}"
        Style="{StaticResource TabItemStyle}"
        Icon="{StaticResource SettingsIcon}"
        ContentTemplate="{DataTemplate SettingsView}" />
</xct:TabView>
```



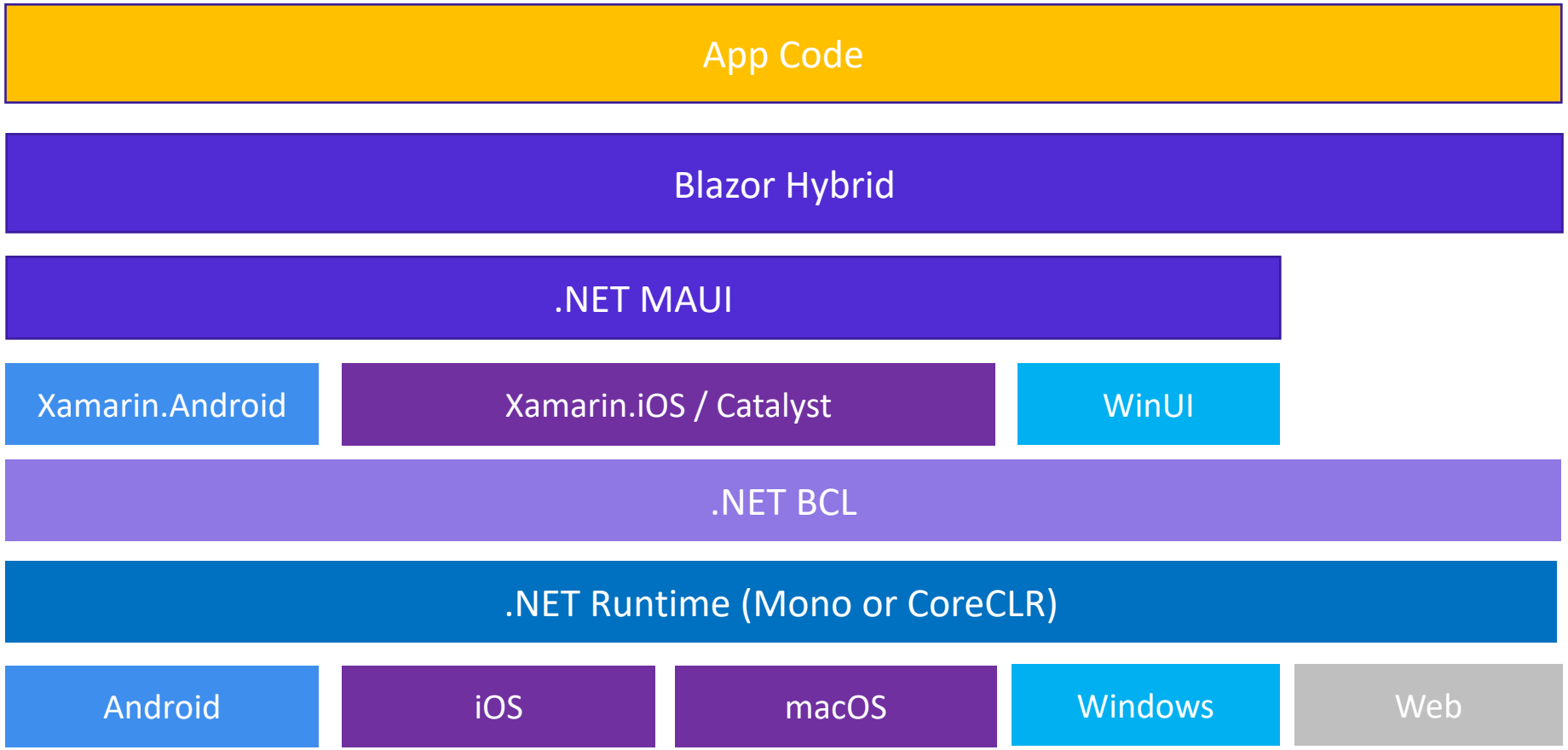
The Future of .NET

.NET MAUI is the evolution of Xamarin.Forms extended from mobile to desktop scenarios with UI controls rebuilt from the ground up for performance and extensibility.

.NET MAUI


App Code
.NET MAUI
Xamarin.Android
Xamarin.iOS / Catalyst
WinUI
.NET BCL
.NET Runtime (Mono or CoreCLR)
Android
iOS
macOS
Windows

.NET MAUI + Blazor



Base Class Library (BCL) Pre-.NET 6

Comparing System.IO.Stream

.NET

```
public abstract void Write(byte[] buffer, int offset, int count);

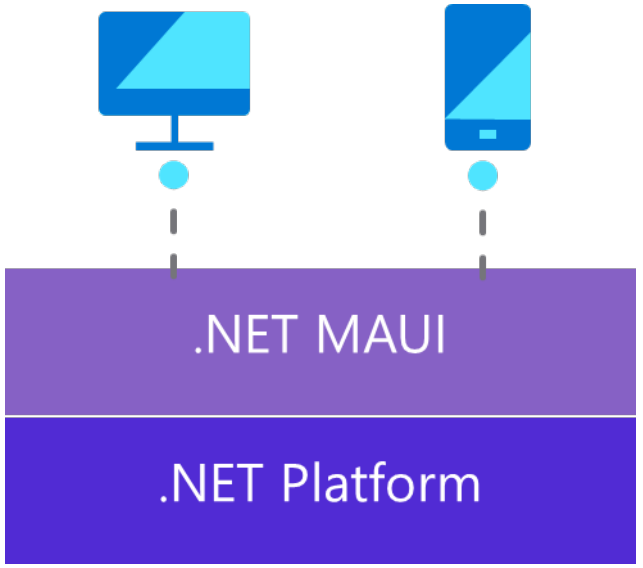
public virtual void Write(ReadOnlySpan<byte> buffer){ ... }
```

Mono

```
public abstract void Write(byte[] buffer, int offset, int count);
```

.NET Multi-platform App UI (MAUI)

- Xamarin integrated into .NET
  - github.com/dotnet/maui
  - github.com/dotnet/android
  - github.com/dotnet/macios
- Common .NET experiences
  - CLI
  - Visual Studio Code
- Android, iOS, macOS, and Windows
- Support for modern app models like Blazor and Comet



EXPERIMENTAL

Goals of .NET MAUI

Quality

12 months of Xamarin quality releases

Reorganized project structure

Standardized class naming and structure

Performance

Cold start <= 1000ms

Responsive UI rendering, animations, and transitions

Reduced view nesting

Design

Simple and elegant default UI theme

Easy control extensibility

Quick cross-platform templating

Platforms

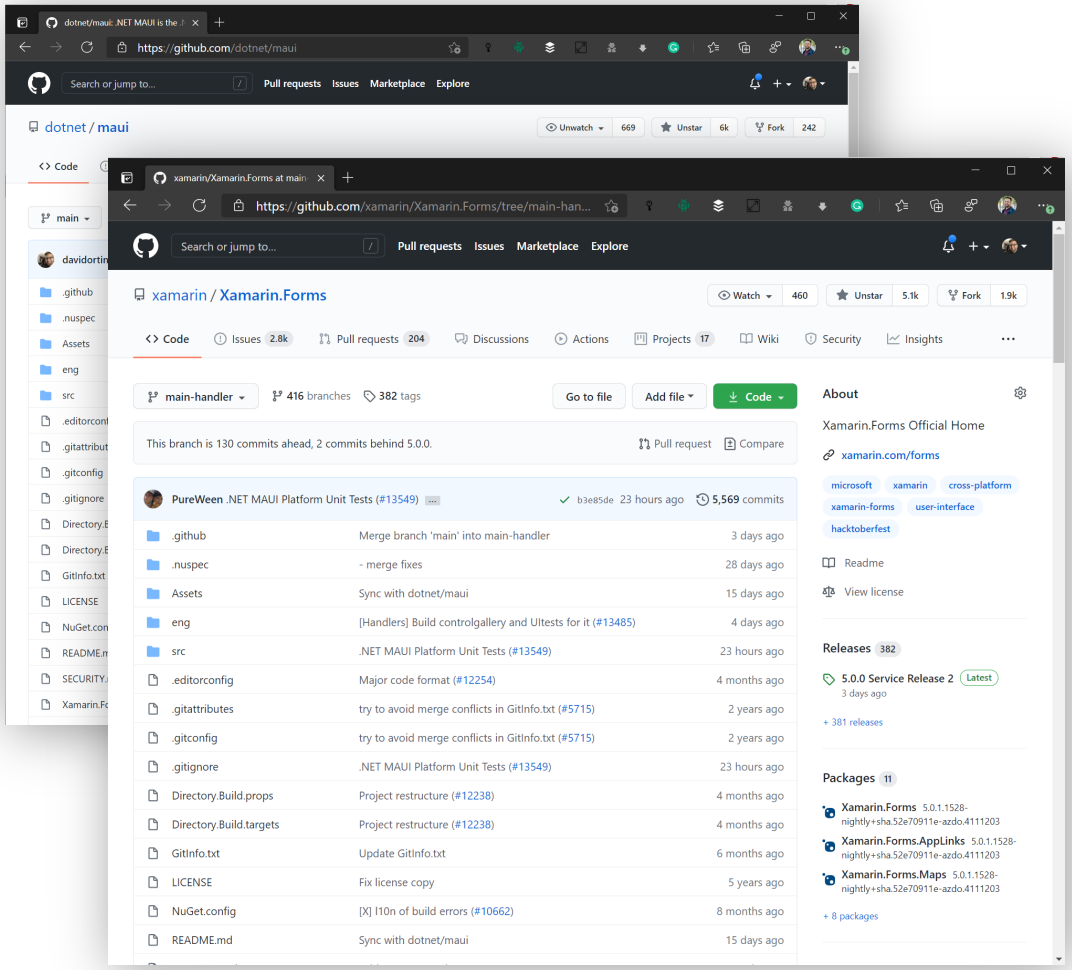
Android

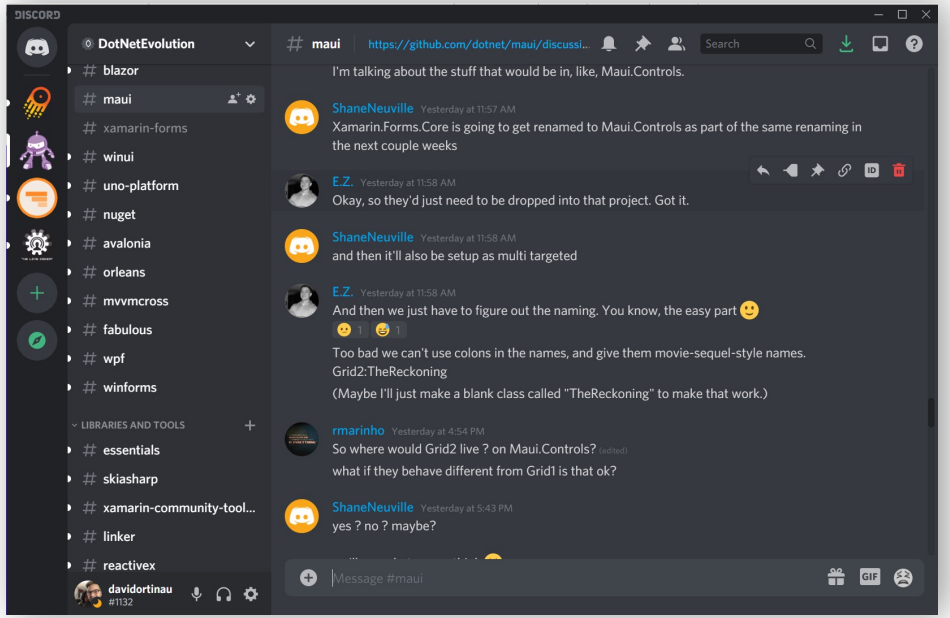
iOS

macOS

Windows

Where is this happening?





https://aka.ms/discord-maui

Open | Closed

- **.NET appeals to net-new developers, students, and new technologists** Theme
  Proposed for .NET 6.0 dotnet/core#5465 opened one month ago
- **.NET has a great client app development experience** Theme
  Proposed for .NET 6.0 dotnet/core#5423 opened one month ago
  - **Blazor developers can target and deploy their applications to desktop platforms, allowing them to run natively** Epic Priority:0 affected-medium area-blazor enhancement severity-blocking
    Proposed for .NET 6.0 dotnet/aspnetcore#27217 opened one month ago Next sprint planning
  - **Desktop developers can package, distribute, release, and update their applications to multiple desktop platforms and architectures** Cost:XL Epic Priority:0
    Proposed for .NET 6.0 dotnet/core#5462 opened one month ago Future rconard
  - **Xamarin developers can upgrade to and use the latest .NET SDKs in their existing applications** Epic Priority:0 Theme: great client app dev XP
    Proposed for .NET 6.0 dotnet/xamarin#2 opened one month ago
    - **.NET developers can run iOS and Android apps on emulators and devices in .NET 6** Cost:XL Priority:0 User Story area-Meta
      Proposed for .NET 6.0 dotnet/runtime#44654 opened 19 days ago steveisok
    - **NuGet packages are updated and available to target .NET 6** Cost:M Priority:1 Theme: great client app dev XP User Story
      Proposed for .NET 6.0 dotnet/xamarin#15 opened 19 days ago
    - **Xamarin developers can acquire and manage mobile .NET SDKs** Cost:M Priority:0 Theme: great client app dev XP User Story
      Proposed for .NET 6.0 dotnet/xamarin#4 opened one month ago
    - **Xamarin developers can leverage a CLI try-convert tool to move basic Xamarin projects to the new project system** Cost:M Priority:1 Theme: great client app dev XP User Story
      Proposed for .NET 6.0 dotnet/xamarin#9 opened one month ago
    - **Xamarin developers can target all supported platforms from a single project, and share resources across them** Cost:XL Priority:0 User Story
      Proposed for .NET 6.0 dotnet/xamarin#11 opened one month ago
    - **Xamarin developers can target Windows with .NET 6** Cost:XL Priority:2 User Story
      Proposed for .NET 6.0 dotnet/xamarin#12 opened one month ago
    - **Xamarin developers can use refactored Xamarin.Core controls for improved performance** Cost:XL Priority:0 User Story
      Proposed for .NET 6.0 dotnet/xamarin#10 opened one month ago
    - **Xamarin developers will experience improved app startup with .NET 6** Cost:L Priority:1 Theme: great client app dev XP User Story
      In Progress for .NET 6.0 dotnet/xamarin#8 opened one month ago
    - **Xamarin developers will share more common developer experiences with .NET developers** Cost:XL Priority:0 Theme: great client app dev XP User Story
      In Progress for .NET 6.0 dotnet/xamarin#7 opened one month ago
    - **Xamarin developers will still be able to develop their legacy projects with Visual Studio** Cost:S Priority:0 Theme: great client app dev XP User Story
      In Progress for .NET 6.0 dotnet/xamarin#6 opened one month ago
    - **Xamarin is integrated with common .NET 6 targets and build commands** Cost:XL Priority:0 Theme: great client app dev XP User Story
      In Progress for .NET 6.0 dotnet/xamarin#16 opened 19 days ago
    - **Xamarin/Blazor developers can target macOS with .NET 6** Cost:L Priority:1 User Story area-Meta
      Proposed for .NET 6.0 dotnet/runtime#44736 opened 19 days ago marek-safar

.NET MAUI Timeline

- Preview 1
  - What's New
    - Runtime and BCL
  - Who should use this release?
    - Contributors
- Preview 2
  - What's New
    - .NET MAUI views, layouts, everything that's been merged
    - Android & iOS will be most complete
  - Who should use this release?
    - Contributors

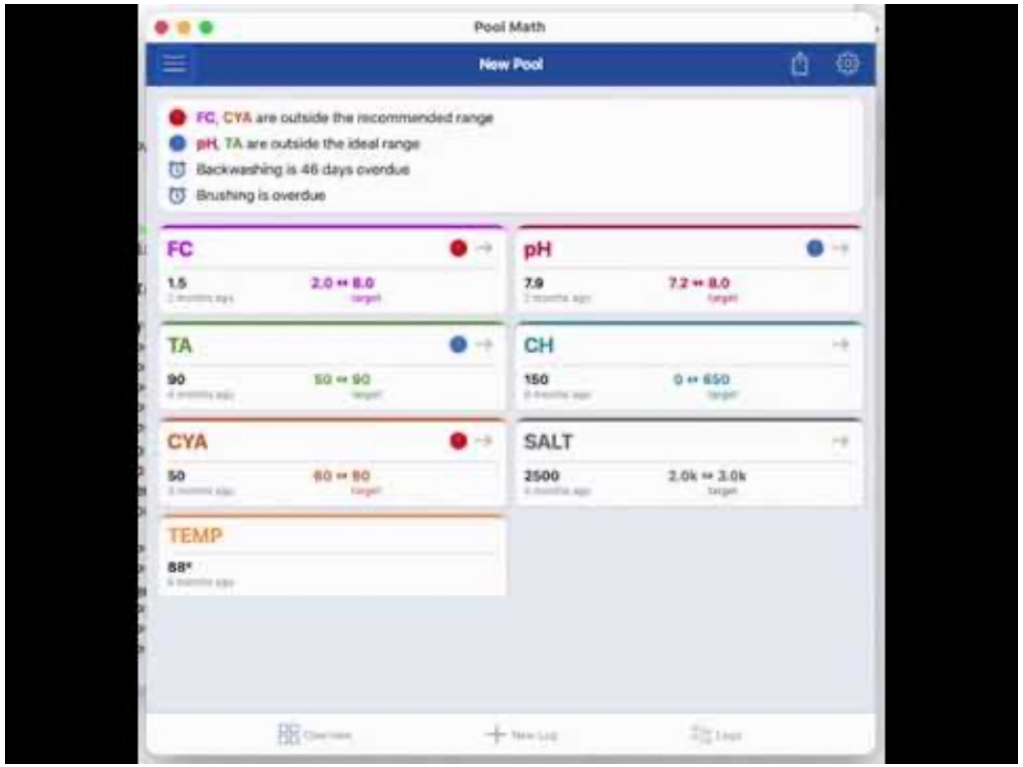
.NET MAUI Timeline

- Preview 3 & 4
  - What's New
    - .NET MAUI will have most controls, layouts, pages, and services
    - Mac Catalyst & WinUI 3
  - Who should use this release?
    - You!
  - What should you do with this release?
    - Build new apps and kick the tires
    - Open issues and join discussion on Discord

.NET MAUI Timeline

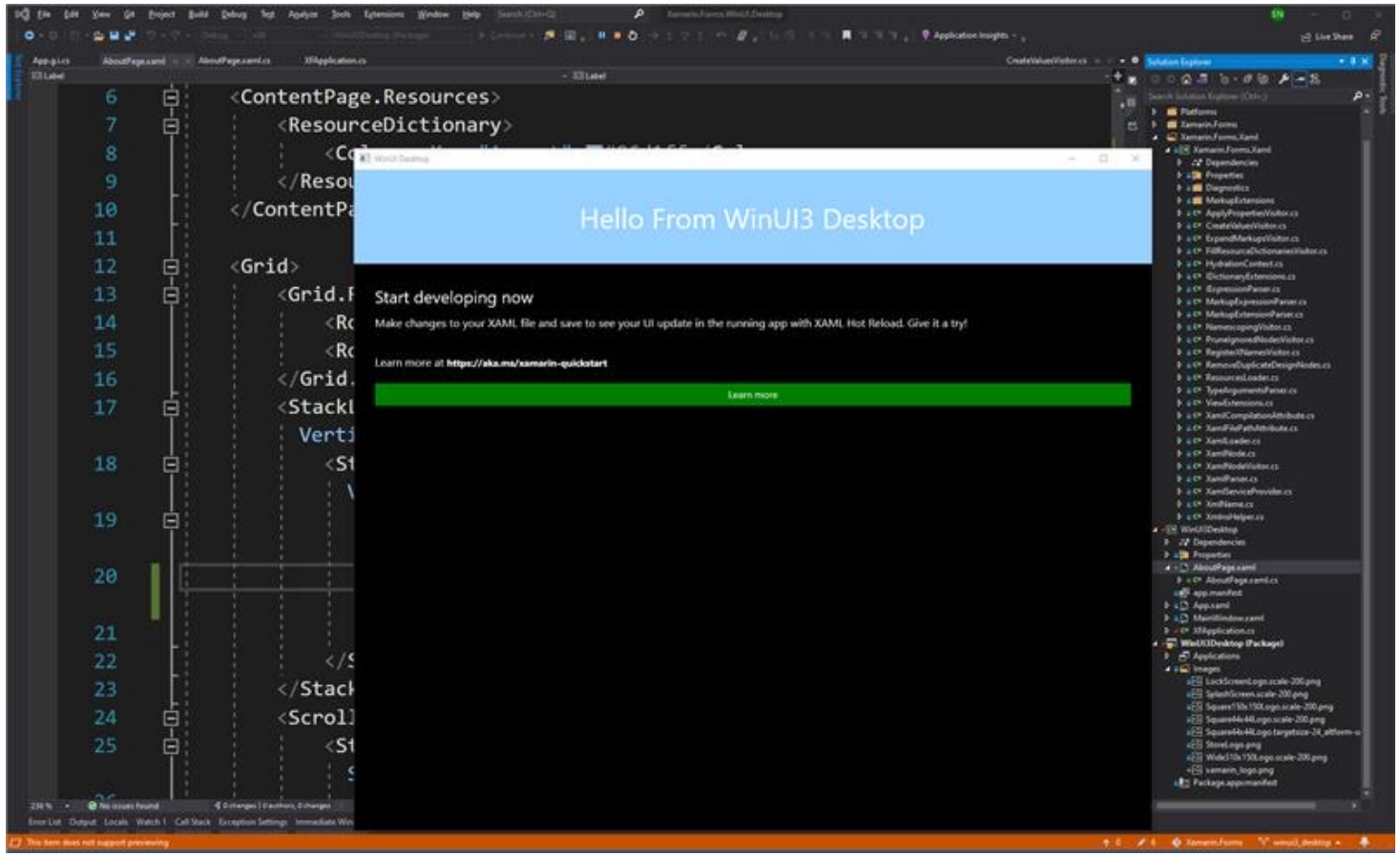
- Preview 5 – Release Candidate
  - What's New
    - .NET MAUI fills the remaining gaps, responds to feedback
  - Who should use this release?
    - Everyone!
  - What should you do with this release?
    - Build new apps and kick the tires
    - Migrate existing Xamarin.Forms apps
    - Open issues and join discussion on Discord

Mac Catalyst



Jonathan Dick – Pool Math
Thanks Frank! https://praeclarum.org/2020/11/10/catalyst.html

WinUI 3



https://github.com/xamarin/Xamarin.Forms/tree/winui3_desktop

OSS Library Maintainer - Monthly Sync

- 1st Tuesday of the Month @ Noon PT / 3 ET
- Teams Call to deep dive on arch and ask ?s
- Current representation from:
  - Prism
  - MvvmCross
  - FreshMvvm
  - ReactiveUI
  - Shiny
  - Xamarin Community Toolkit
  - Mobile Build Tools

To join:
david.ortinau@microsoft.com

Thanks for joining!

